EMNLP 2026

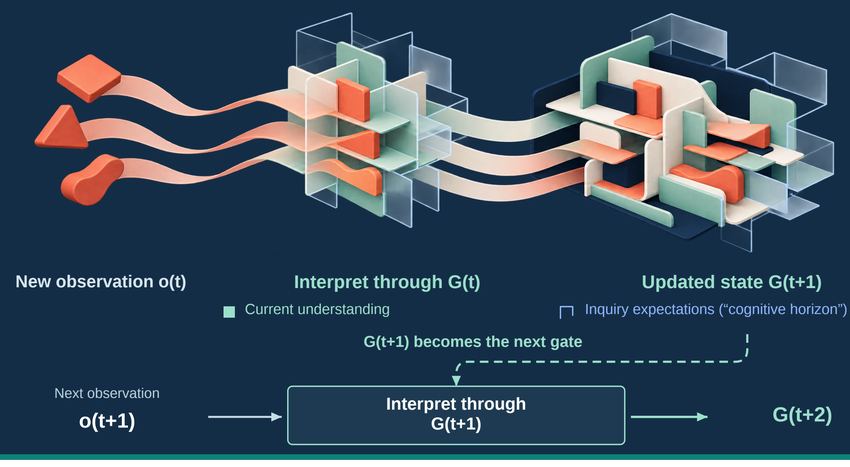
# VeriTrace

## Evolving Mental Models for Deep Research Agents

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## A The research question

### How does an agent build and revise understanding through interaction?

Persistent artefacts carry information across search rounds. VeriTrace regulates how new evidence changes concepts, findings, progress and inquiry, making the mental model an explicit control state.

## B A cognitive graph regulates the research trajectory

**Cognitive graph G(t) = (S(t), C(t))**

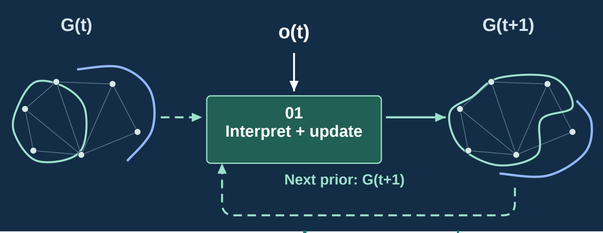
**S(t) / structure**

Concepts · dependencies
Protected user dimensions

**C(t) / content and progress**

Criteria · findings · quality
UNKNOWN / PARTIAL / KNOWN
KNOWN: core criteria satisfied
Edge goals · history · resolution

Current understanding · Inquiry expectations (“cognitive horizon”)




compiled state

candidate signals

**Planner**

Select unresolved edge; instantiate task criteria.

task + criteria

**strategy**

**Searchers**

Parallel retrieval with local context.

pages

**Reader**

Extract findings; score each page.

persistent mismatch

**02 Deviation feedback**
**δ → explicit search strategy**

observation o(t)

new frame

**03 Schema revision**

S(t) → S(t+1); retain evidence and constraints.

**01 Interpretive update**

Information status · Cognitive status
Knowledge associations

Interpret o(t) through prior G(t); produce C(t+1).

**Reassess what is understood, what remains uncertain, and what further inquiry may resolve.**

Prior G(t) anchors interpretation

**Assimilation: update understanding and progress**
**G(t+1) gates o(t+1) in the next round.**

Mechanical evidence records

**G(T) + evidence store**
**No UNKNOWN / no TO-SOLVE, or budget**
**Quote · criterion · URL · task ID**

→ **Outline** — Node-linked sections → **Section plan** — 2–5 evidence IDs / insight → **Writer** — Prose with bound citations → **Cited report** — Traceable evidence

Solid / observations, actions, updates · Dashed / prior-state context · Dotted / direct evidence records

### 01 / Interpretive update

**METACOGNITIVE MONITORING**

Prior concepts and criteria anchor where evidence belongs and how it changes the state.

**Information status**

| | |
|---|---|
| **Criterion-satisfying** | Integrate into findings |
| Redundant | Deduplicate |
| Contradictory | Retain evidence; flag conflict |
| Unexpected | Route to a dedicated channel |

**Cognitive status · Concepts and inquiry goals**

Recompute concept resolution; carry unmet criteria into later tasks.

**Knowledge associations**

Link new findings to prior concepts and evidence.
Route relevant findings across nodes.
Support integration and inform possible reframing.

**A coherent state supports hypothesis checks and corrective reframing.**

### 02 / Deviation feedback

**PREDICTIVE PROCESSING / Direct inquiry**

**δ = (CR, AAP, φ, ψ)**

CR: evidence–need alignment; AAP: credibility (1–5).
φ: blocked (0/1); ψ: unexpected strength (1–4).

| | |
|---|---|
| Blocked | **SUBSTITUTE** |
| High CR + high AAP | **EXPLOIT** |
| High CR + low AAP | **VERIFY** |
| Low CR + surprise | **PIVOT** |
| Otherwise | **EXPLORE** |

Priority order. High ≥ 3.5; surprise = moderate or stronger (Appendix VI).

**Typed mismatch yields a targeted next action.**

### 03 / Schema revision

**ACCOMMODATION / Change the frame**

Persistent deviation or accumulated contradictions trigger revision of concepts, relations, goals and criteria.

| | |
|---|---|
| **R_conc** | Expand an abstract concept into concrete items |
| **R_aug** | Add a missing, relevant concept and connect it |
| **R_pivot** | Group inquiry dimensions under each concrete item |
| **R_prune** | Prune irrelevant, unprotected nodes with no findings |
| **R_correct** | Revise affected downstream goals and criteria |

Preserve findings* and user dimensions; resume inquiry on the revised graph.

**Search and restructuring alternate.**

* Documented noise-item cleanup exception.

## C Evidence from reported and matched configurations

### Reported configurations / DRB

| | OVERALL / 0–60 | Overall | Insight |
|---|---|---|---|
| **Proprietary systems** | | | |
| Claude Deep Research | | 45.00 | 42.79 |
| OpenAI Deep Research | | 46.45 | 43.73 |
| Gemini 2.5 Pro DR | | 49.71 | 49.45 |
| Doubao Deep Research | | 44.34 | 40.56 |
| Kimi Researcher | | 44.64 | 41.97 |
| **Open systems / reported backbones** | | | |
| LangChain ODR / GPT-5 | | 50.60 | 50.76 |
| EnterpriseDR / Gemini 2.5 Pro | | 50.62 | 51.24 |
| WebWeaver / Qwen3-235B-A22B | | 50.62 | 51.00 |
| RhinoInsight / Gemini 2.5 Pro | | 50.92 | 51.45 |
| FS-Researcher / Claude Sonnet 4.5 | | 53.94 | 55.85 |
| **VeriTrace configurations** | | | |
| **Config-3B** | | **51.96** | **55.22** |
| **Config-27B** | | **52.60** | **56.12** |
| **Config-Plus** | | **53.50** | **57.08** |
| **Config-DeepSeek** | | **55.77** | **59.56** |

Table 2; DRB n = 100. Different backbones, as reported.
RACE scores ×100; 50 = expert-reference parity.

### Matched Qwen3.5-27B

| System | DRB Overall | DRB Insight | DeepConsult Win rate |
|---|---|---|---|
| Enterprise-DR | 49.16 | 50.17 | 64.7% |
| FS-Researcher | 48.03 | 47.05 | 72.8% |
| WebWeaver | 50.77 ± 0.34 | 51.30 ± 0.33 | 75.2% |
| **VeriTrace** | **52.60 ± 0.31** | **56.12 ± 0.26** | **81.1%** |

**+1.83** Overall points

**+4.82** Insight points

**+5.9** Win-rate points

Gains vs WebWeaver. DRB n = 100; means ± SD over 3 full runs for VeriTrace and WebWeaver. DeepConsult n = 102; Gemini 2.5 Pro judge (Tables 3–4).

## D Which mechanisms matter?

Full: 52.60 ± 0.31. A2: R_aug / R_prune only. A3: naive text concatenation.

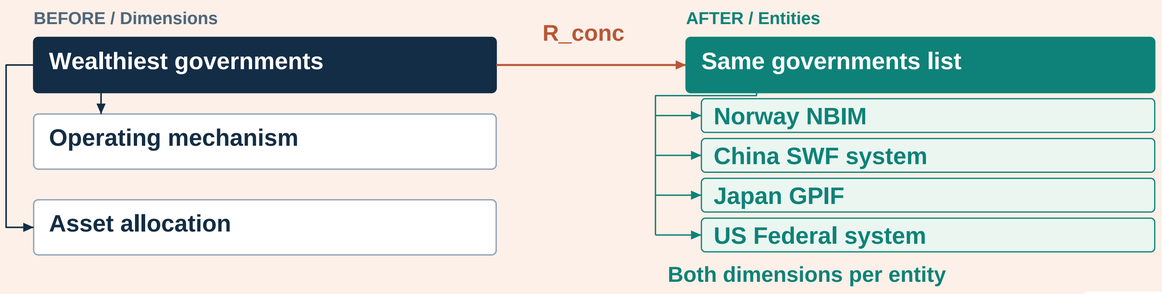
| Ablation | Overall | Δ (pp) | Search | 90-min |
|---|---|---|---|---|
| A1 − Deviation feedback | 50.52 | **−1.96** | 1.22× | 86/100 |
| A2 Restricted revision | 51.28 | **−1.36** | 0.98× | 96/100 |
| A3 − Interpretive update | 50.75 | **−1.83** | 0.41× | 99/99 |
| A4 − Graph topology | 51.39 | **−1.20** | 1.14× | 99/99 |
| Afull All four removed | 49.80 | **−2.81** | 0.54× | 100/100 |

Δ: paired per-query change vs 3-run Full. Search: call ratio.
90-min: within-budget completion (Table 5).

Frequent revision (p ≥ 2/3): A2 −4.61 (n=9), A3 −4.87 (n=11). p = fraction of Full runs with R_conc / R_pivot / R_correct (Table 6).

## E Reframing in a logged trajectory

Query 53: evidence grouped operating mechanism and asset allocation by investing entity (Appendix XV).




**8 prospective task slots → 4 inquiries; 13 findings preserved.**

hz496@cam.ac.uk · github.com/WolfGaga/VeriTrace · Per-task state; cognitive science used as a design vocabulary.